UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (10 MARKS)

Answer ALL questions.

1. Which loop is guaranteed to execute its body at least once?
   A. for
   B. while
   C. if
   D. do...while

2. What is displayed by the following code?

```
int x = 7;
cout << x / 2 << " " << x % 2;
```

   A. 3 0
   B. 3 1
   C. 3.5 1
   D. 4 1

3. Which structure chooses between two alternative paths?
   A. for
   B. struct
   C. array
   D. if...else

4. Which statement immediately exits a loop?
   A. break
   B. continue
   C. return 0
   D. cin

5. A computer system's hardware includes which of the following?
   i. input devices
   ii. output devices
   iii. central processing unit
   iv. main memory
   v. secondary storage
   A. i, ii, iii, v
   B. i, ii, iii, iv, v
   C. i, ii, iii, iv
   D. i, iii, iv

6. Which phase of the program development life cycle defines what the program must do?
   A. Analysis
   B. Testing
   C. Coding
   D. Maintenance

7. Given the declaration, what is displayed?

```
char word[6] = {'p', 'r', 'i', 'n', 't', '\0'};
cout << word[3];
```

   A. p

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B. i
C. t
D. n

8. Which declaration creates an integer array of five elements?
A. int a(5);
B. array a[5];
C. int a[5];
D. int a;

9. Which of the following are valid identifiers?
i. total_score
ii. 2total
iii. _count
iv. int
A. ii and iv
B. iii and iv
C. i and ii
D. i and iii

10. Which declaration creates a floating-point variable named price?
A. float price;
B. char price;
C. bool price;
D. int price;

PART B (60 MARKS)

QUESTION 1

This question tests the program development life cycle.

a) Define source code and object code.

(2 marks)

b) Describe THREE stages of the program development life cycle.

(3 marks)

QUESTION 2

A concert charges the ticket prices shown in Table 1. A negative age is invalid.

Table 1. Concert Ticket Prices

Age Range	Ticket Price (RM)
0 - 12	15.00
13 - 17	25.00
18 and above	40.00

a) Draw a flowchart for the ticket price selection.

(8 marks)

b) Write pseudocode and a complete C++ program for the ticket price selection.

(5 marks)

QUESTION 3

This question tests operator precedence and assignment.

a) Evaluate the following expressions using the declarations in the code block.

```
int a = 12, b = 5;
double c = 2.5;
```

(6 marks)

b) Write C++ statements to declare two integers a and b, read their values, and display their sum, difference, product and remainder.

(5 marks)

QUESTION 4

Study the following nested loop:

```
for (int i = 1; i <= 3; i++) {
    for (int j = 1; j <= i; j++) {
        cout << j;
    }
    cout << endl;
}
```

a) Trace the program and show the exact output.

(10 marks)

b) Identify TWO logic errors in a version that prints four rows but reuses j without resetting it. Rewrite the corrected code.

(5 marks)

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QUESTION 5

A shop stores the item prices shown in Table 1.

a) Declare an array named prices using the values in Table 1. Display all values.

Table 2. Item Prices

Item	Price (RM)
1	20.50
2	55.00
3	15.90
4	75.00
5	10.00

(6 marks)

b) Write a program segment that applies a 10% discount to prices above RM50 and stores the results in a second array. Display both arrays.

(10 marks)

PART C (30 MARKS)

QUESTION 1

A lecturer records the marks of ten students in an array.

Write a complete C++ program that reads ten marks, displays the average, the highest mark and its index, and counts how many marks are above the average.

(15 marks)

QUESTION 2

A bookstore records up to five books. Table 1 shows an example of the fields.

Table 3. Book Record Fields

Title	Price (RM)	Quantity
Book A	45.00	3
Book B	30.00	5
Book C	55.00	2

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

END OF QUESTION PAPER

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